

ABIRAMI R

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PROFILE SUMMARY

Developer & Machine Learning Engineer with expertise in **Python, Java, ReactJS, SQL**, Skilled in building **end-to-end applications** and developing **ML models using TensorFlow, Keras, PyTorch, and Scikit-learn**, with a strong foundation in **Exploratory data analysis**.

SKILLS

Analytical tools: Spice simulation, Git, Microsoft excel.

Languages known: Python, HTML, CSS, Javascript, Java.

Frameworks / technologies: ReactJS, Bootstrap, Tailwind CSS, Flask, FastAPI, TensorFlow, Keras, PyTorch, Scikit-learn, NLP, Pandas, NumPy, MySQL, MongoDB, Microcontrollers (Arduino, ESP32), Embedded System, REST APIs, fine-tuning LLMs and Hugging Face Transformers.

EXPERIENCE

CSIR – National Aerospace Laboratories - Bengaluru, Karnataka.

Aug 2025-Present

ML Project Associate

- Developed an acoustic source localization framework using MFC/PZT rosette sensors and wavelet-based Lamb wave amplitude extraction by complex data science methods.
- Estimated wave direction of arrival through dynamic strain tensor analysis and multi-rosette triangulation.
- Transformed processed directional data **into physics-informed boundary conditions** for a PINN-based inverse localization model.
- Achieved **~98% localization accuracy** over **10mm MSE**, outperforming traditional RNN-based models in complex, anisotropic structural environments.

CONTRIBUTIONS

Google Summer of Code 2025, Google.

Apr 2025

Student developer program – Contributor (Accord project)

- Migrated template playground's legacy CSS codebase to Tailwind CSS to enhance scalability, maintainability, and development speed.
- Refactored core UI components like navbar, buttons, forms, and modals using Tailwind's utility-first classes.
- Reduced CSS bundle size and eliminated redundant styles, **improving overall performance and load times by ~25%**.
- Collaborated with maintainers to get PRs merged, including hover interaction fixes and consistent styling patterns.

PROJECTS & RESEARCH

SatDecay PINNs - Python – Git - Pytorch.

2025

- Built a hybrid physics + ML framework to improve **satellite re-entry prediction** for Low Earth Orbit objects.
- Combined **SGP4 orbital propagation** with data-driven error correction using historical TLE and decay datasets.
- Engineered Gradient Boosting and **PINN models to capture atmospheric drag** variability and nonlinear orbital decay.
- Delivered a computationally efficient pipeline suitable for real-time debris tracking and operational forecasting systems.

lot-bedside control panel Arduino - Embedded systems - GIT.

2025

- Designed and developed a **smart IoT-based bedside control panel** equipped with sensors for temperature, gyroscope, and heart rate monitoring, improving monitoring coverage **from 60% to 100%**.
- Implemented **automatic SMS alerts** via a GSM module to notify caregivers in case of abnormal heart rate, reducing average response time **from 10 minutes to under 6 minutes (70% to 90%)**.
- Click here : [lot-control-panel](#)

EDUCATION

B.E. in Electronics and Communication Engineering.

2021-2025

DMI College Of Engineering, Chennai

- CGPA: 77.59%

+2 PCM.

2020-2021

St. Columban's A.I.HR.SEC School, Chennai

- GPA: 84%

LICENSES & CERTIFICATIONS

Great-Learning – Python for ML - Python fundamentals - Generative AI.

2024

GUVI-HCL – Machine Learning 101 – Learnathon.

2024